

Macroeconomic Theory and Policy: Lecture W10 (Ch.15 and 18)

Quantitative DSGE Models

- Full DSGE models incorporate the complete dynamic response of all economic variables to all shocks
- The Smets–Wouters (2007) model is used by macroeconomic forecasters and central banks around the world as a building block
 - ▶ Incorporates shocks previously discussed
 - ▶ Includes both sticky prices and sticky wages and more
- One of the key advantages of this type of model is that it can be estimated
 - ▶ By “estimated,” we mean that the parameters of the DSGE model are chosen to make the model match the data for the U.S. economy as accurately as possible

Shocks

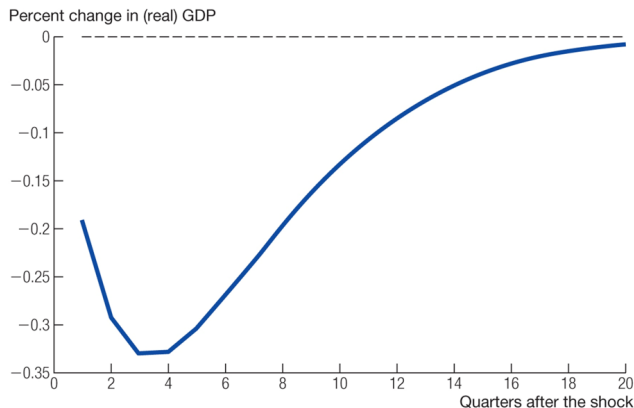
- Quantitative DSGE models incorporate multiple shocks simultaneously
- The Smets-Wouters model features:
 - ▶ Productivity shocks
 - ▶ Shocks to government purchases
 - ▶ Financial frictions
 - ▶ Investment shocks
 - ▶ Labour market shocks
 - ▶ Shocks to markups in product markets

Impulse Response Functions

- Impulse Response Functions (IRF) is one of the main tool to analyze the DSGE model (and multivariate time series model)
- The IRF shows how one macroeconomic variable of interest responds over time to an economic shock
- Each graph shows how each variables responses to a temporary shock
- The next slide shows the IRF for GDP in the estimated Smets-Wouters model
 - ▶ By what percent does GDP change after a temporary 1 percentage point increase in the fed funds rate?

The Response of GDP to a Monetary Policy Shock

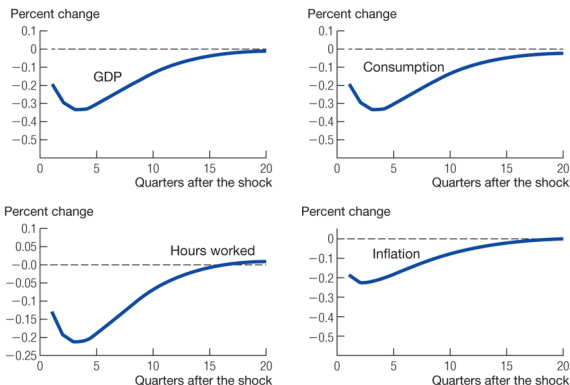
- A 1 % increase in the fed funds rate causes GDP to fall by about 0.2 % in the first quarter after the shock
- At its peak three to four quarters after the shock, GDP is lower by 0.33 %



Source: Frank Smets and Rafael Wouters, "Shocks and Frictions in U.S. Business Cycles: A Bayesian DSGE Approach," *American Economic Review*, vol. 97, no. 3 (2007), pp. 586–606.
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The Dynamic Effects of a Monetary Policy Shock

- The DSGE models include many other variables, like consumption, hours worked, and inflation, allow us to study how each of these variables responds to a shock
- The response of the economy looks very much like the response we saw in our AS/AD framework

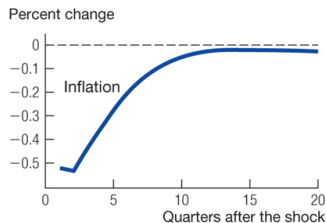
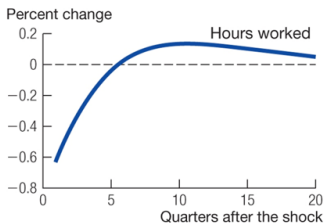
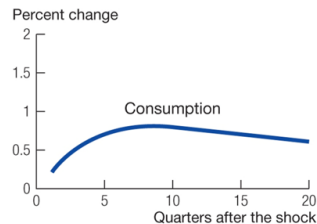
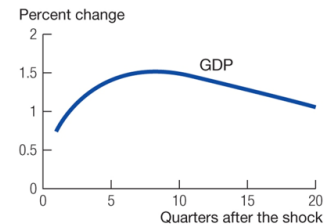


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The Dynamic Effects of a Monetary Policy Shock

- When there is a surprise tightening of monetary policy (a rise in interest rate), there are less investment
- As a result, less output to be produced which lower the consumption
 - ▶ Given the permanent income hypothesis, the consumption should varies less than the output
 - ▶ But Smets-Wouters's substantial adjustment costs to investment causes consumption to move by more (if the output is not being used in investment, it would go toward consumption)
 - ▶ Yet, a good model should not imply that the consumption move more than output
- As the output fall, according to the PC, the inflation rate should fall

The Dynamic Effects of a TFP Shock



Source: Frank Smets and Rafael Wouters, "Shocks and Frictions in U.S. Business Cycles: A Bayesian DSGE Approach," *American Economic Review*, vol. 97, no. 3 (2007), pp. 586–606.
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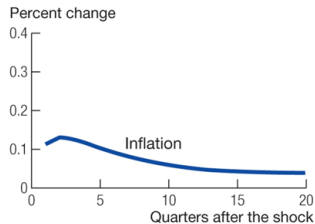
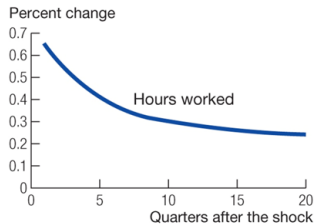
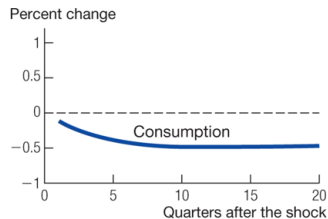
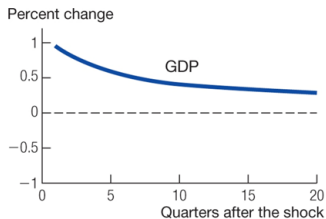
The Dynamic Effects of a TFP Shock

- The TFP shocks considered here decay over time instead of disappear entirely in one period
- As the positive TFP shocks arrive, the GDP rises
 - ▶ The increase is even higher than the the increase in the TFP (i.e., 1%)
 - ▶ This is because of the capital accumulation, which is similar to the case discussed in the Solow model
- Given the permanent income hypothesis, individual prefer to smooth consumption.
 - ▶ The consumption increases less than the increase in output

The Dynamic Effects of a TFP Shock

- The hour worked may look puzzling, as the stylized model would suggest an increase in both wage and employment
 - ▶ This model feature sticky prices, the firm supply as many as amount demanded
 - ▶ Since the consumption rises by little, the rise in TFP is more than enough to supply the amount demanded in the economy
- The TFP shocks also raises potential GDP and expands supply in the economy
 - ▶ In the AS/AD model, this shifts down the AS curve
 - ▶ However, the actual output doesn't increase as much as the shock initially, the actual output would be lower than the potential output temporary
 - ▶ Resulting a fall in inflation rate

The Dynamic Effects of a Shock to Government Purchases

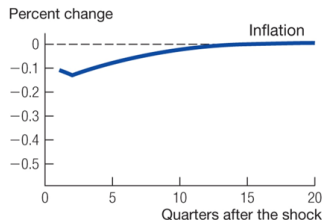
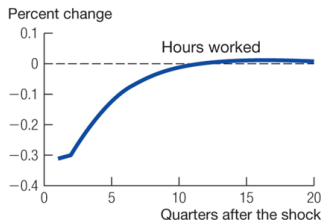
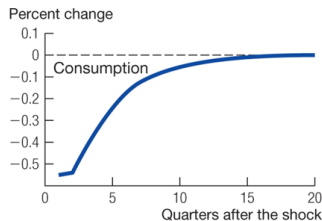
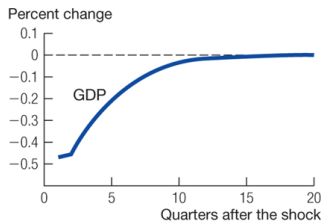


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The Dynamic Effects of a Shock to Government Purchases

- The stylized model suggests that the wages would fall while the hours worked increases
 - ▶ As the government increases its spending, consumer would expect an increase in tax at some point
 - ▶ The wealth decrease and lead to a decrease in consumption
 - ▶ The labour supply shift to the right
- The model generate similar response as expected
- Since the labour supply increases, the effect on inflation is little

The Dynamic Effects of a Financial Frictions Shock



Source: Frank Smets and Rafael Wouters, "Shocks and Frictions in U.S. Business Cycles: A Bayesian DSGE Approach," *American Economic Review*, vol. 97, no. 3 (2007), pp. 586–606.
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The Dynamic Effects of a Financial Frictions Shock

- This particular financial shock implemented as a wedge between the rate of saving and borrowing, because of asymmetric information problem
 - ▶ Implies it has the largest effect on consumption
- Otherwise, the shock looks very similar to a monetary policy shock, as both work through interest rates

Conclusion

- Macroeconomists argue about the importance of various shocks and various frictions in accounting for economic fluctuations
 - ▶ These arguments often center around the extent to which certain frictions are quantitatively important
 - ★ How sticky are wages and prices?
 - ★ How important are credit market frictions?
 - ★ How elastic is labor supply in the economy?

Conclusion

- The basic framework we should use to understand macroeconomic fluctuations developed over the past 50 years
 - ▶ A key element is a basic growth model like the Solow model
 - ▶ Including more microfoundation elements
- The basic DSGE framework has been enriched in many ways recently
- The models make precise quantitative predictions about the complete dynamics of a host of endogenous variables

What are You Going to Learn (Ch. 18)

- Government spending, taxation, budget deficits, and the debt-GDP ratio nationally and globally
- The government's intertemporal budget constraint
- The economic consequences of budget deficits
- The fiscal problem of the twenty-first century: how to finance rising health expenditures

Introduction

- The government can finance expenditures by
 - ▶ using tax revenue
 - ▶ printing money
 - ▶ borrowing
- But the government's budget must balance in present value
 - ▶ The amount the government spends must equal the amount the government earns in terms of present value
 - ▶ Budget deficits today must be offset by budget surpluses in the future
- The “fiscal problem of the twenty-first century”
 - ▶ If current policies remain unchanged, forecasts show the U.S. and many other countries' deficits and debt-to-GDP ratios will explode

U.S. Government Spending and Revenue

- In 2018, government spending in the United States was
 - ▶ \$6.7 trillion
 - ▶ 20.0 percent of GDP
 - ▶ more than \$20,500 per person
- Tax revenues were 16.2 percent of GDP

The U.S. Federal Government Budget, 2018

TABLE 18.1

The U.S. Federal Government Budget, 2018

	Percentage of GDP	Dollars per person
Total expenditures	20.0	12,550
Health (including Medicare)	5.6	3,480
Social Security	4.8	3,020
National defense	3.1	1,930
Income security	2.4	1,510
Net interest	1.6	990
Other	2.6	1,620
Total revenues	16.2	10,170
Individual income taxes	8.2	5,140
Social insurance and retirement receipts	5.7	3,580
Corporate income taxes	1.0	630
Other	1.3	830
Budget deficit	-3.8	-2,380

Source: *Economic Report of the President*, 2018, Table B-19.
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The U.S. Federal Government Budget, 2018

- It is expected that the expenditure healthcare as a fraction of GDP will increase over time, because:
 - ▶ Federal entitlements are becoming more generous
 - ▶ A growing proportion of the population qualifies for benefits
 - ▶ With rising income, expenditure on healthcare is expected to increase at a faster rate than consumption

Government Spending and Revenue

- The budget balance: The difference between tax revenues and spending
- A budget surplus: Tax revenues $>$ Spending
- A budget deficit: Tax revenues $<$ Spending
 - ▶ The government must borrow by selling bonds
- A balanced budget: Tax revenues $=$ Spending

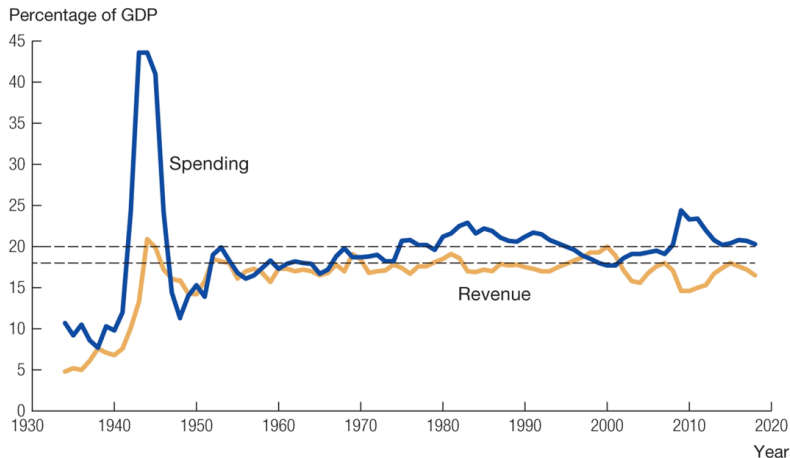
Government Spending and Revenue

- In 2018, there was a budget deficit equal to 3.8 percent of GDP
 - ▶ The government does not usually sell assets to finance the deficit
 - ▶ Usually, the government borrows from lenders within the United States and abroad
- When the government sells bonds, they use the money as a loan
 - ▶ Suppose a 10-year bond issued in 2021 pays \$100 when it matures in 2031
 - ▶ In 2021, the investor may pay \$55 for this bond (and that amount is the loan to the government)
 - ▶ The remaining \$45 the government pays is the nominal interest
- In 2018, the government issued more than \$750 billion in new bonds (>\$2,300 per person)

Spending and Revenue over Time

- After World War II, spending and tax revenues rose to about 18 percent of GDP
 - ▶ The two were roughly the same fraction of the GDP
- In 1970, systematic budget deficits began to emerge
 - ▶ Deficits as a fraction of GDP peaked in 1983
 - ▶ There was a brief surplus in 1998, followed by another deficit in 2002
- In 2009, the budget deficit was around 10 percent of GDP
 - ▶ During the recession, tax revenues fell and government spending increased

U.S. Federal Government Revenue and Spending



Source: *Economic Report of the President*, 2019, Table B-46.
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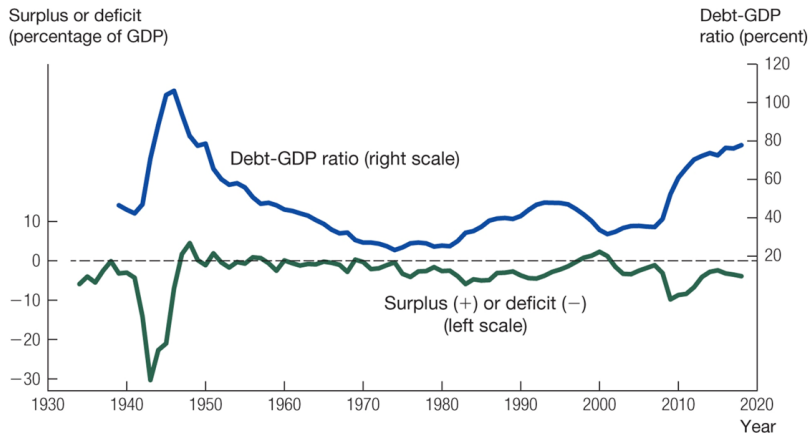
The Debt-GDP Ratio

- Government debt: outstanding stock of bonds that have been issued in the past
- In 2005, the debt-GDP ratio was just over 35 percent
- In 2018, the debt-GDP ratio was nearly 80 percent, a huge increase
- About half of the debt in recent years is owed to foreigners

The Debt-GDP Ratio

- The government itself also holds a large amount of bonds
 - ▶ The Social Security trust fund that will be paid to future retirees held \$2.7 trillion in assets in 2014
 - ▶ One branch of government loaning to another has no affect on government “wealth.”
 - ★ The asset and debt created offset one another

Federal Debt and Deficits in the United States

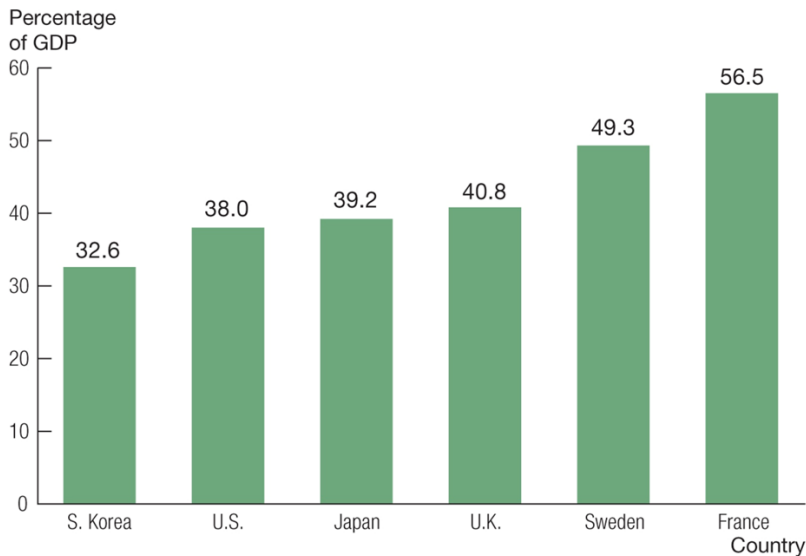


Source: *Economic Report of the President*, 2019, Table B-46.
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International Evidence on Spending and Debt

- Among the richer OECD countries, the United States has a lower than average
 - ▶ government spending-to-GDP ratio
 - ▶ debt-GDP ratio
- Norway is an exception among many rich countries
 - ▶ Negative debt-GDP ratio
 - ▶ Norway benefits from large government-owned petroleum and natural gas reserves, which boost its revenues and lead to budget surpluses
 - ▶ Knowing these revenues will be depleted, the Norwegian government saves these surpluses as assets in a Government Petroleum Fund to finance future government spending

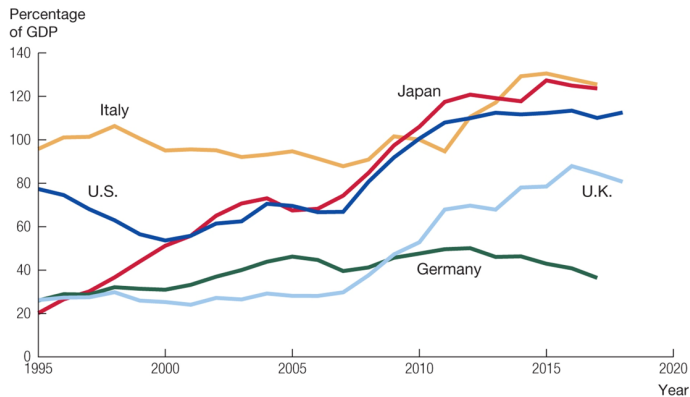
Government Spending around the World, 2017



Source: <http://data.oecd.org/gga/>.
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Debt-GDP Ratios around the World

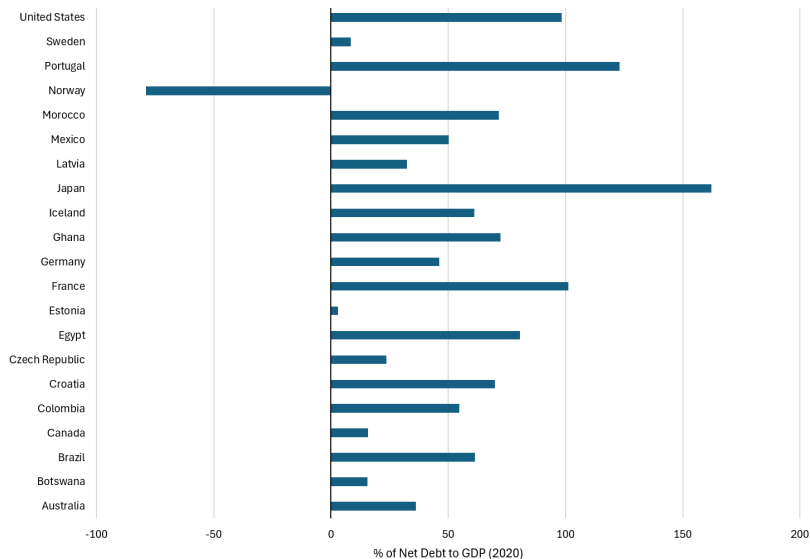
- In Japan, the ratio has risen sharply since 1995, from a low of about 20 percent to a high of more than 120 percent in 2018



Source: <http://data.oecd.org/gga/>.
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Debt-GDP Ratios around the World

- The net debt to GDP ratio varies a lot across country. But, higher number does not always mean a country is at risk. It depends on many country specific factor



Debt-GDP Ratios around the World

- The Debt-GDP ratio comprises two components: Debt and GDP
- If the GDP kept growing faster than debt, the debt-GDP ratio could decrease
- When assessing the risk of a country defaulting, it's necessary to project the future growth of the country
 - ▶ Many factor could come into place as we learn in previous chapter

The Government Budget Constraint

- The flow version of the government budget constraint holds in each period
- The sources of funds to the government must equal the uses of funds

$$G_t + Tr_t + iB_t = T_t + \Delta B_{t+1} + \Delta M_{t+1}$$

where G_t is the gov't purchases, Tr_t is the transfer payment, iB_t is the interest payments on debts, T_t is the taxes, ΔB_{t+1} is the New Borrowing and ΔM_{t+1} is the change in money stock

- ΔB_{t+1} and ΔM_{t+1} can be negative

The Government Budget Constraint

- Assume for this chapter

- ▶ The change in the stock of money is zero (ΔM_{t+1})
- ▶ Transfer payments are zero ($T_t = 0$)

$$B_{t+1} = (1 + i)B_t + G_t - T_t$$

- LHS is the stock of debt at the start of next year
- RHS is the Stock of debt this year, including the required interest payments plus the budget deficit
- The primary deficit: $G_t - T_t$
 - ▶ If the $G_t - T_t = 0$, then the growth of B is $g_B = \frac{B_{t+1}}{B_t} - 1 = i$
- The total deficit: $G_t + iB_t - T_t$

Next Lecture

- Next lecture will finish Ch.18